

Domenico Valentino

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EDUCATION

McGill University | B.Eng., Computer Engineering | Expected May 2030

Dawson College | Mathematics and Computer Science | Honour Roll | May 2026

TECHNICAL SKILLS

Languages: Rust, C/C++, Java, TypeScript/JavaScript, Python, Swift, SQL

Tools: React, Tauri, React Native, Node.js, Firebase, Git, Linux

Systems: Audio DSP, concurrency, Core ML, WPILib, sensor integration

SELECTED ENGINEERING EXPERIENCE AND PROJECTS

OverCue | DJ software and firmware | Jul 2026 - Present

overcue.gg | Publicly released software and firmware

- Built core Rust/Tauri and React components of a macOS/Windows app for DJs to prepare music libraries and separate tracks into vocal and instrument stems; integrated Swift/Core ML inference.
- Developed ARM64 Linux C/C++ audio extensions with atomic synchronization, bounded buffers, resampling, and click-suppressed playback transitions.
- Added DJM knob control of CDJ stems with switching back to standard EQ; external apps handle stem processing.
- Tested audio processing for concurrency, cancellation, malformed inputs, and loudness limits.

FIRST Robotics | Programming Lead and Robotics Mentor | 2022 - 2025

- Developed competition-deployed Java/WPILib robot software integrating autonomous driving, vision-based localization, and coordinated scoring mechanisms.
- Built TechScout and TechInsights with React Native/Firebase for offline scouting, QR data transfer, and scoring/cycle-time analysis; shipped TechScout across three seasons (App Store).
- Led 10 programmers across teams 3990 and 9406; delegated tasks, reviewed pull requests, and taught beginners who progressed to writing robot control systems.

Team 3990 award (2024): Long Island Autonomous Award.

2025 Championship match video

Kaskaraa Instruments | Co-Founder and Software Lead | Apr 2025 - Present

kaskaraa.com | District 3 incubation | Mitacs-supported research

- Lead software development for prototype automated microtomes that prepare thin tissue sections for pathology, including UI, motor control, safety logic, and AI imaging.
- Develop custom embedded platforms, circuit boards, and wiring, integrating electronics with instrument software.
- Manage interns and collaborate with ~10 Concordia graduate engineering students.

ADDITIONAL ENGINEERING PROJECT

6502 Computer | Digital hardware and assembly | Summer 2026

- Built and programmed a working 6502 computer; wired CPU, memory, clock/reset, address decoding, and I/O, then brought up the system using ROM programming and assembly.

ADDITIONAL EXPERIENCE

Apple | Product Specialist | Oct 2025 - Present

- Explain product capabilities and system behavior in plain language; serve as a technical resource for colleagues troubleshooting customer questions.
 - De-escalate difficult customer interactions by explaining product behavior and the reasons behind recommendations.
- English and French (fluent); Italian and Spanish (conversational).